TEAM G-1

PRESENTED BY :

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OMNICHANNEL RETAIL SALES AND INVENTORY ANALYTICS PROJECT

2026 2026 2026 2026 2026 2026 2026

DATA ANALYTICS

TEAM G-1

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DATA UNDERSTANDING

Omnichannel Retail Sales and Inventory Dataset

This project analyzes an AI generated dataset of omnichannel retail business by integrating data across customers, orders, products, inventory, and warehouses.

Data Scope

The dataset consist of 2023-2024 retail transactions (50K rows) across 51 branch store and 10 warehouses in India.

Data Relation

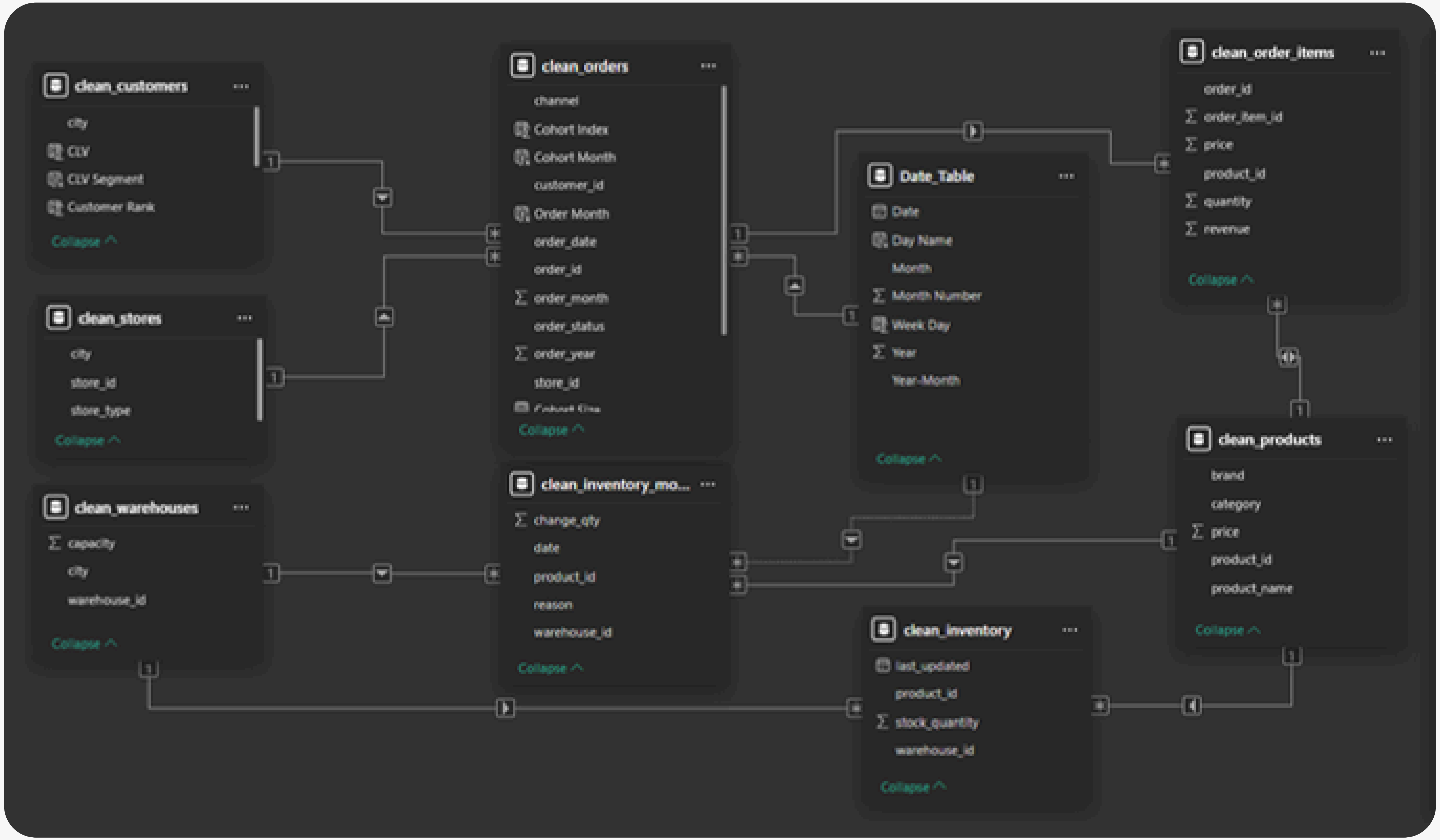
The dataset consist of 8 related tables including : customers, orders, order_items, products, inventory, inventory_movements, stores, and warehouses

Project Goal

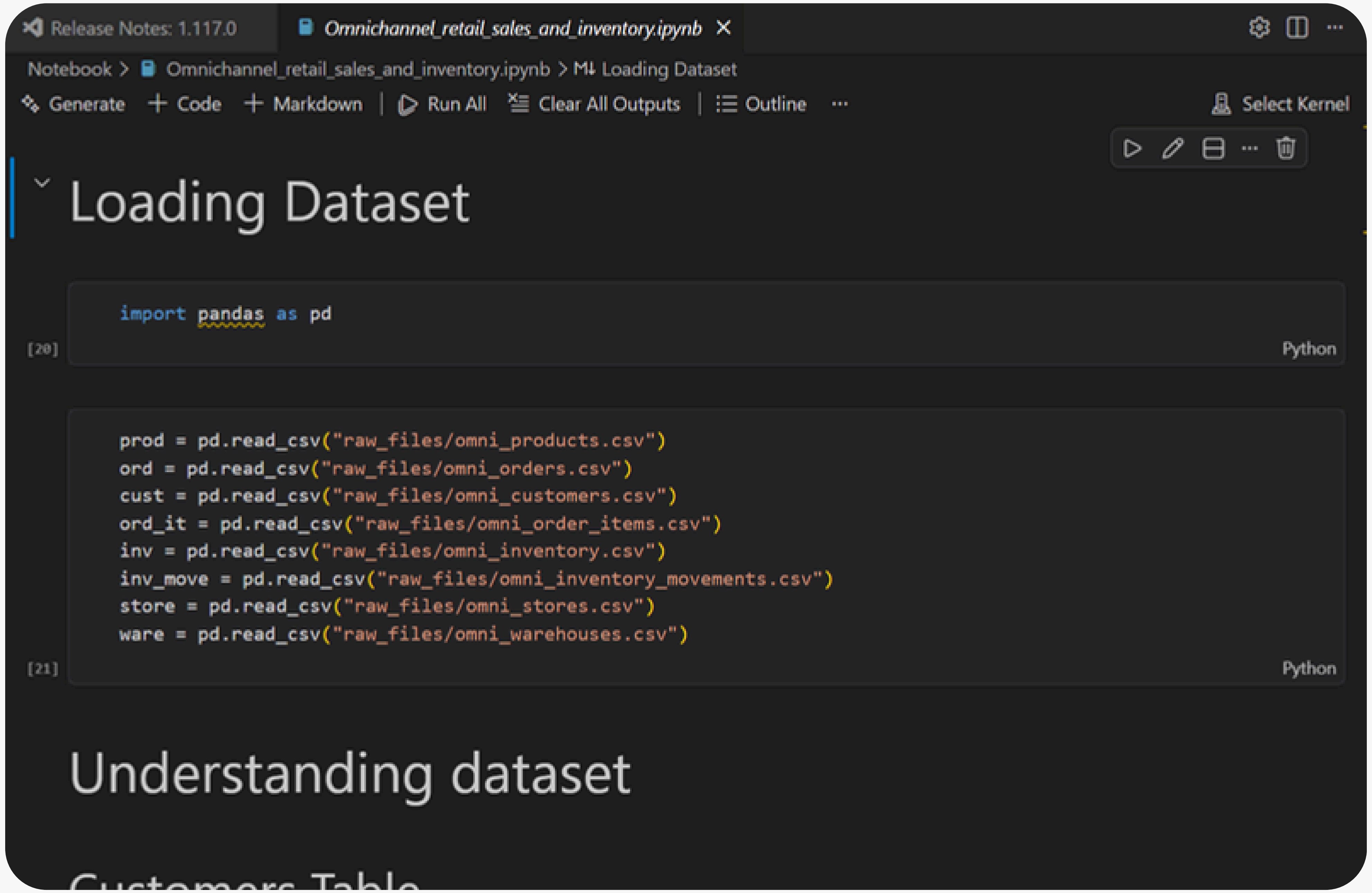
- Understand sales performance across channels (Store, Online, App)
- Track inventory movements and stock levels
- Analyze customer behavior and retention
- Build a scalable data model for business intelligence

DATA DIAGRAM

DATA DIAGRAM



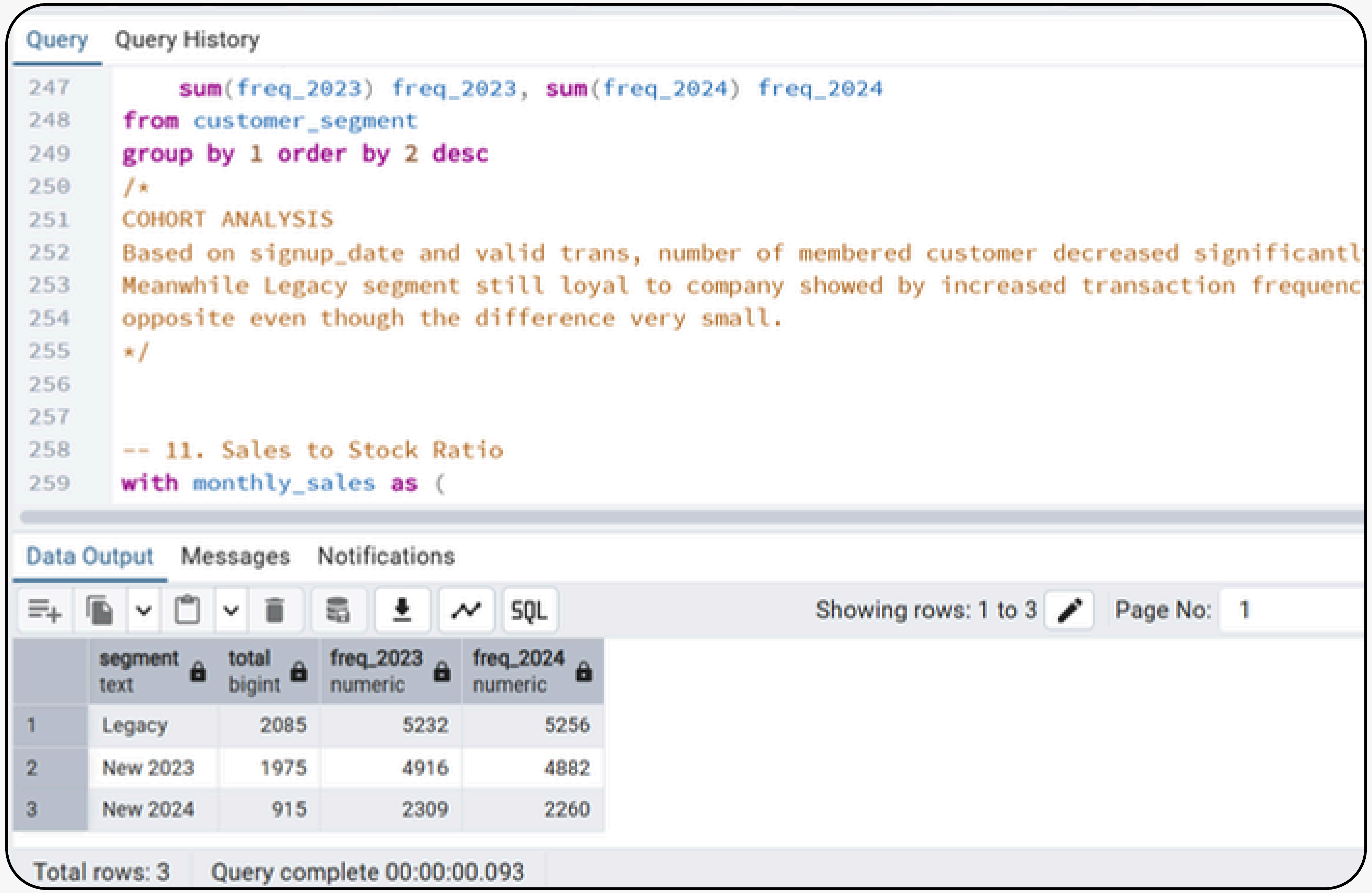
PYTHON DATA CLEANING



- Standardized column names and its data types
- Removed duplicates and invalid values
- Converted date formats
- Handled missing values (e.g. online store mapping)
- Fixed inventory movement logic (positive/negative quantities)
- Created derived columns (revenue, order_year, order_month)
- Primary and foreign key validation
- etc

PANDAS **NUMPY** **MATPLOTLIB** **SEABORN** **SQLALCHEMY** **PANDAS** **NUMPY** **MATPLOTLIB**

POSTGRESQL DATA ANALYSIS



Query Query History

```sql
    sum(freq_2023) freq_2023, sum(freq_2024) freq_2024
from customer_segment
group by 1 order by 2 desc
/*
COHORT ANALYSIS
Based on signup_date and valid trans, number of membered customer decreased significantl
Meanwhile Legacy segment still loyal to company showed by increased transaction frequenc
opposite even though the difference very small.
*/


-- 11. Sales to Stock Ratio
with monthly_sales as (
```

Data Output Messages Notifications

SQL Showing rows: 1 to 3 Page No: 1

	segment text	total bigint	freq_2023 numeric	freq_2024 numeric
1	Legacy	2085	5232	5256
2	New 2023	1975	4916	4882
3	New 2024	915	2309	2260

Total rows: 3 Query complete 00:00:00.093

- Revenue Over Time
- Best Sales Product over Time
- Cancelled and Returned Product over Time
- Product Performance by Revenue
- Store and City Performance by revenue
- Online Channel Comparison by Revenue
- Basket Analysis by Average Order Value
- Customer Loyalty by Order Count and Revenue Generated
- Customer Distribution and Its Revenue Generated
- Cohort Analysis
- Sales to Stock Ratio
- Inventory Turn Over Ratio

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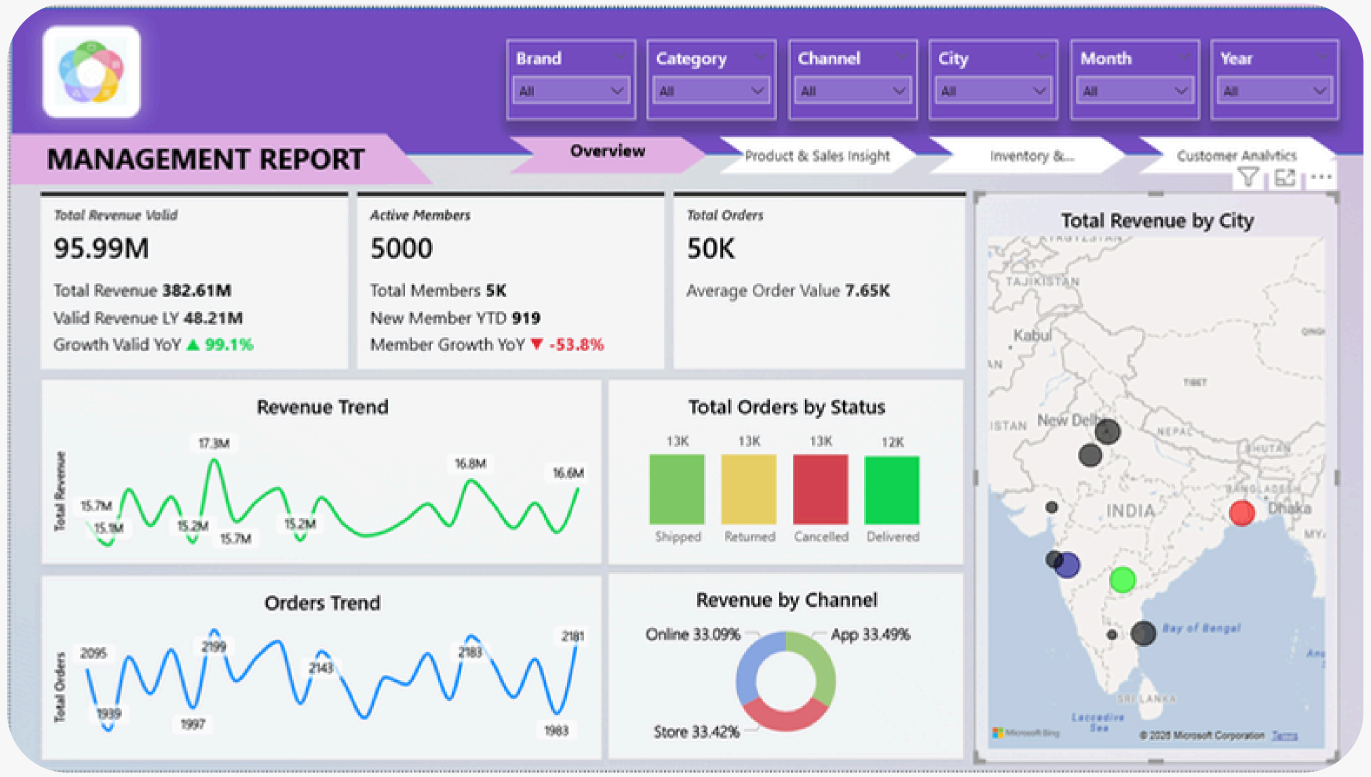
POWER BI DASHBOARD

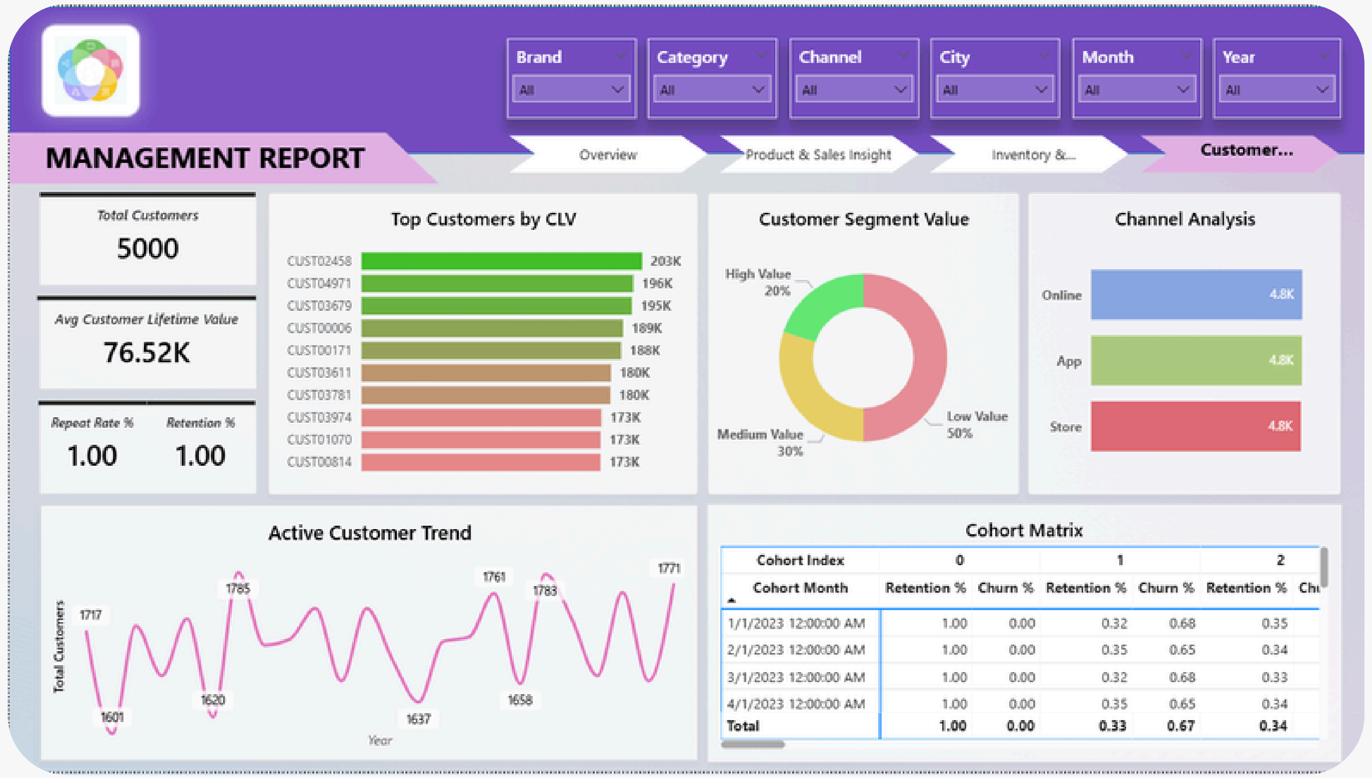
OVERVIEW



PRODUCT AND SALES INSIGHT

INVENTORY AND OPERATION

CUSTOMER ANALYSIS



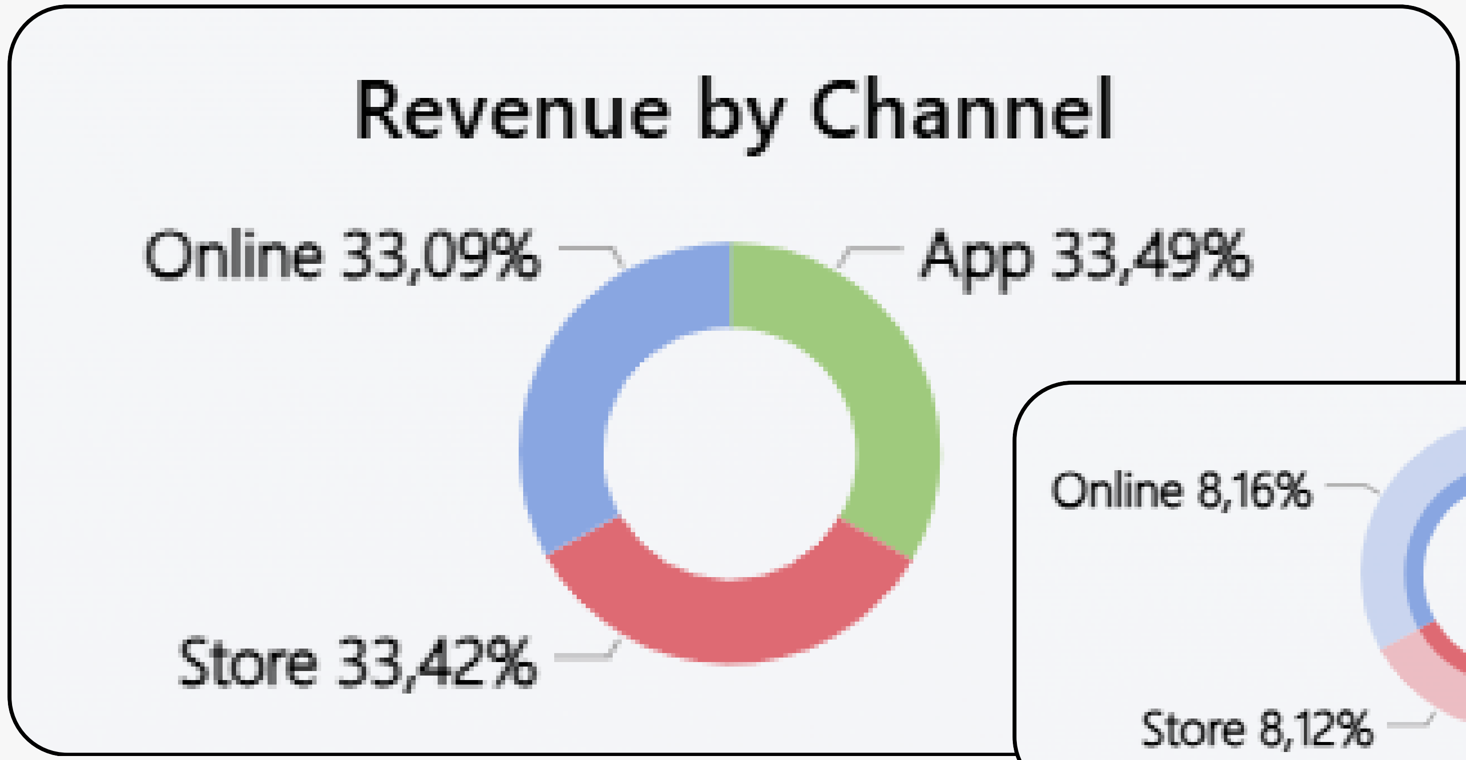
KEY INSIGHT

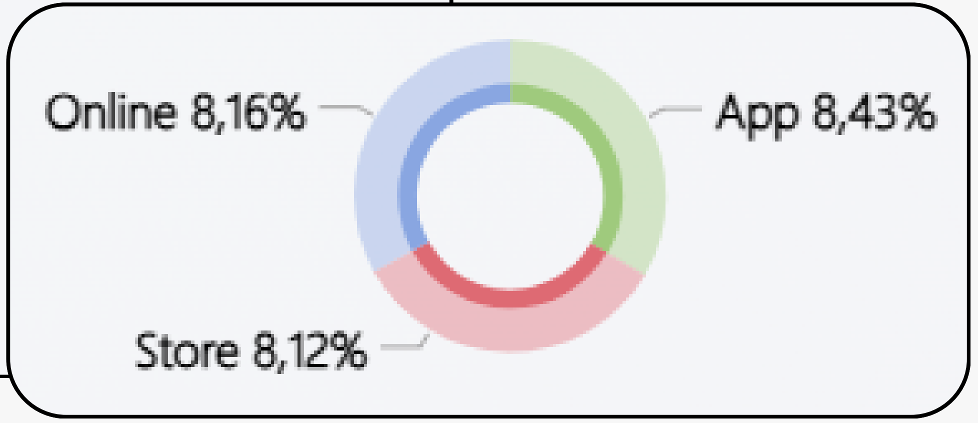
DISCLAIMER

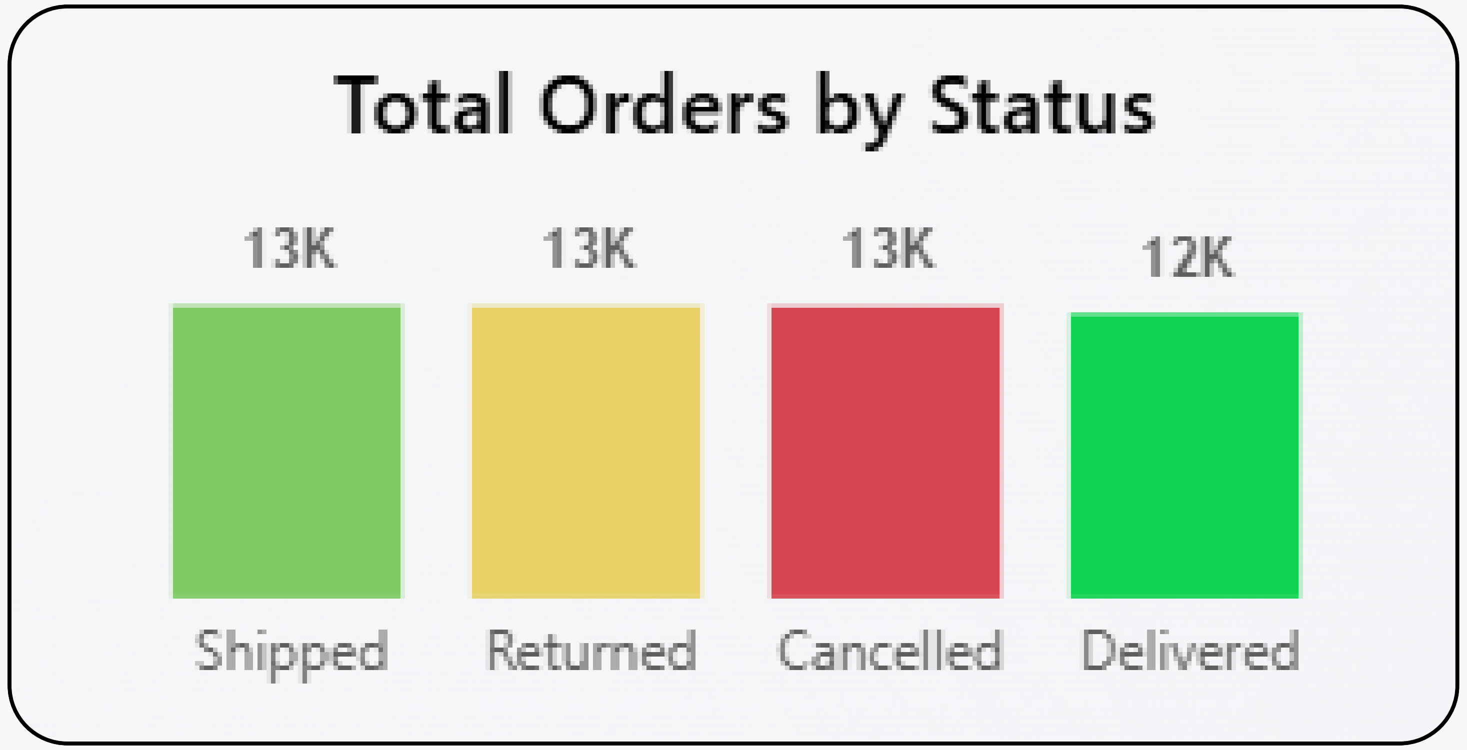
All metrics shown in these slides are based on a synthetic AI dataset. Numerical extremes are reflective of artificial modeling rather than real-world conditions. Results are for presentation and framework review purposes only.

REVENUE AND CHANNEL

STRONG DIGITAL DEMAND MASKED BY CRITICAL ORDER FULFILLMENT ISSUES

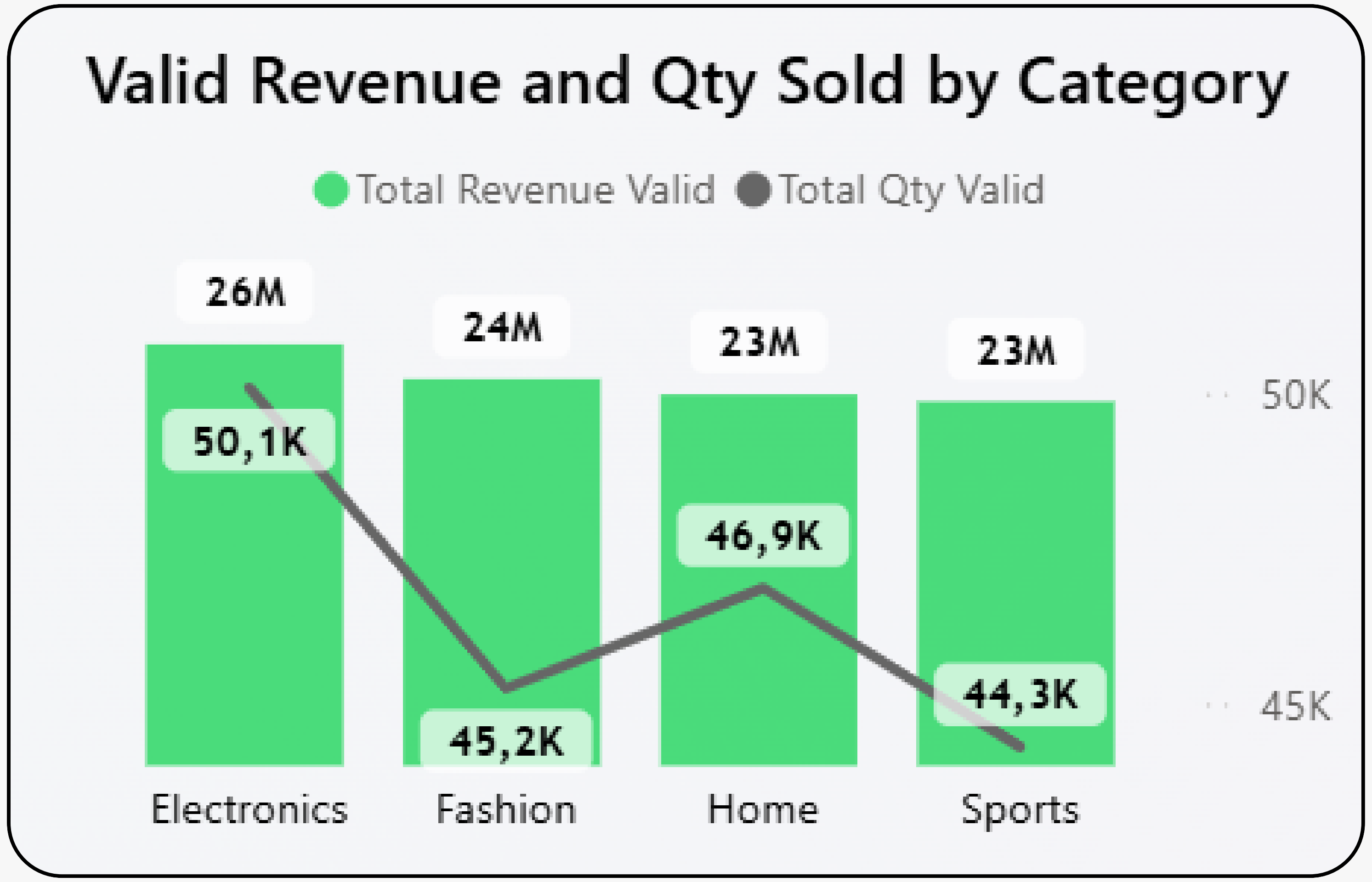






Despite **strong digital traction with Online and App channels** contributing over 66% of total revenue, the high volume of Cancelled and Returned orders signals a critical bottleneck in the fulfillment process. **The "Delivered" revenue represents only a fraction of potential earnings** up to 8.43% each of total Revenue, suggesting that while the digital storefront effectively captures interest, operational inefficiencies or product quality issues are causing significant profit leakage. Highlighting an urgent need to **improve order success rates** to ensure sustainable growth.

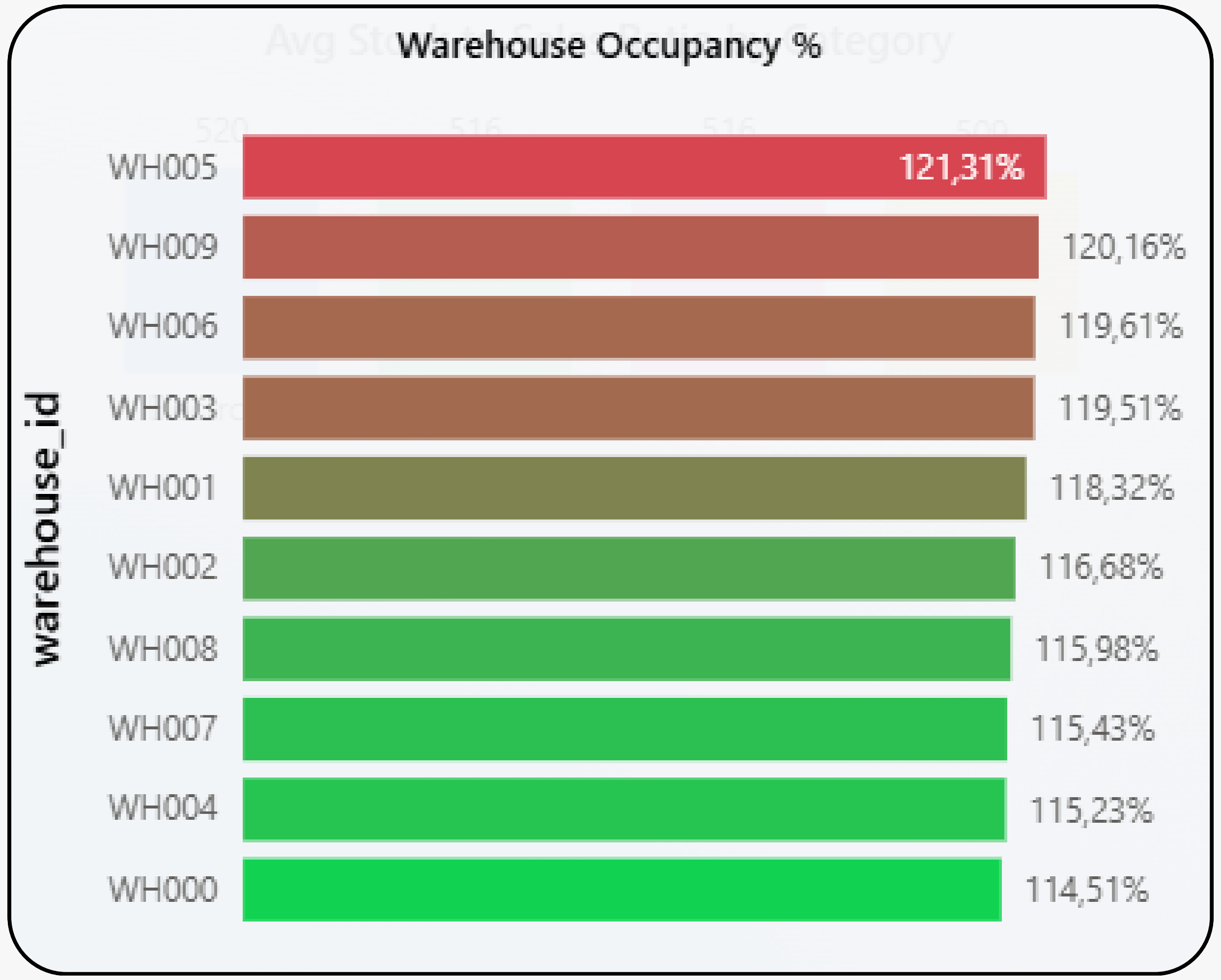
SMALL VOLUME, MASSIVE IMPACT: FASHION'S HIGH-VALUE ADVANTAGE



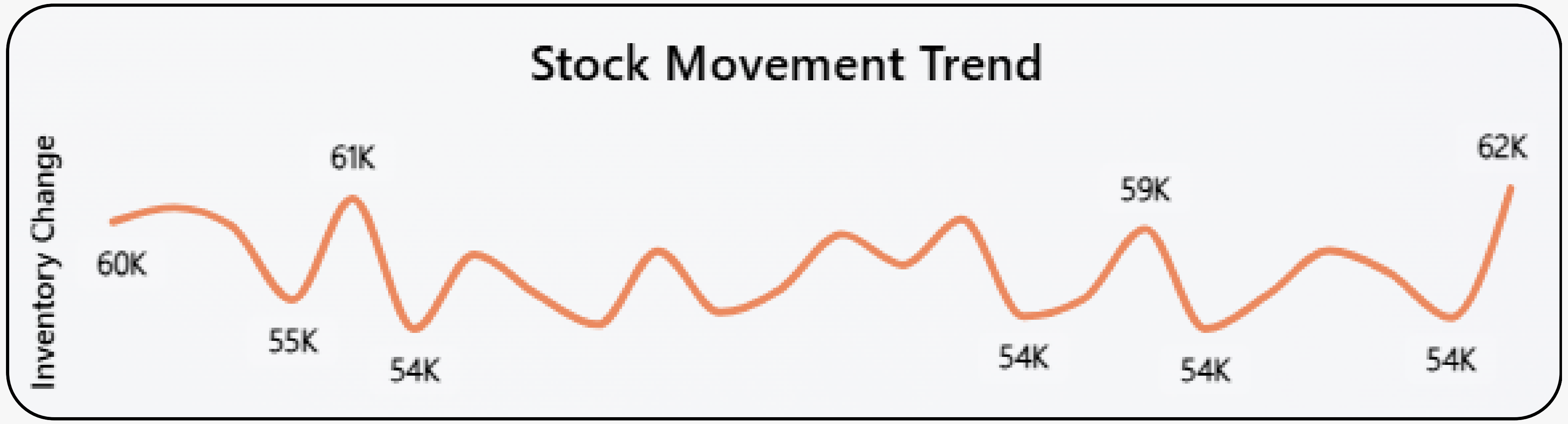
The **Fashion category** is a masterclass in efficiency, securing a **staggering 24M in revenue while moving significantly fewer units** 45.2K compared to other categories. It effectively outearns the higher-volume Home segment despite selling nearly $2,000$ fewer items, showcasing a superior revenue-per-unit ratio. This highlights Fashion as **a premium engine for the business**, proving that high-value transactions can drive top-tier financial performance without relying on sheer sales volume.

INVENTORY AND OPERATION

CRITICAL OVERCAPACITY AMID UNSTABLE INVENTORY INFLOWS



The organization is currently facing a severe storage crisis, with **every warehouse exceeding its physical capacity**, peaking at 121.31%. While the Stock Movement Trend shows active sales dips to 54K, these gains are immediately erased by **aggressive restocking spikes** that recently surged to 62K. This "yo-yo" pattern suggests that inventory is being forced into facilities that have no room to breathe. Without an **immediate pause on new intakes or an urgent expansion of storage space**, the supply chain remains at a breaking point, risking both operational efficiency and workplace safety.



INVENTORY AND OPERATION

RECLAIMING FROZEN CASH FLOW BY RESOLVING CRITICAL INVENTORY STAGNATION

515.37

AVG Sales To Stock Ratio

0.26

AVG Inventory Turn Over Index

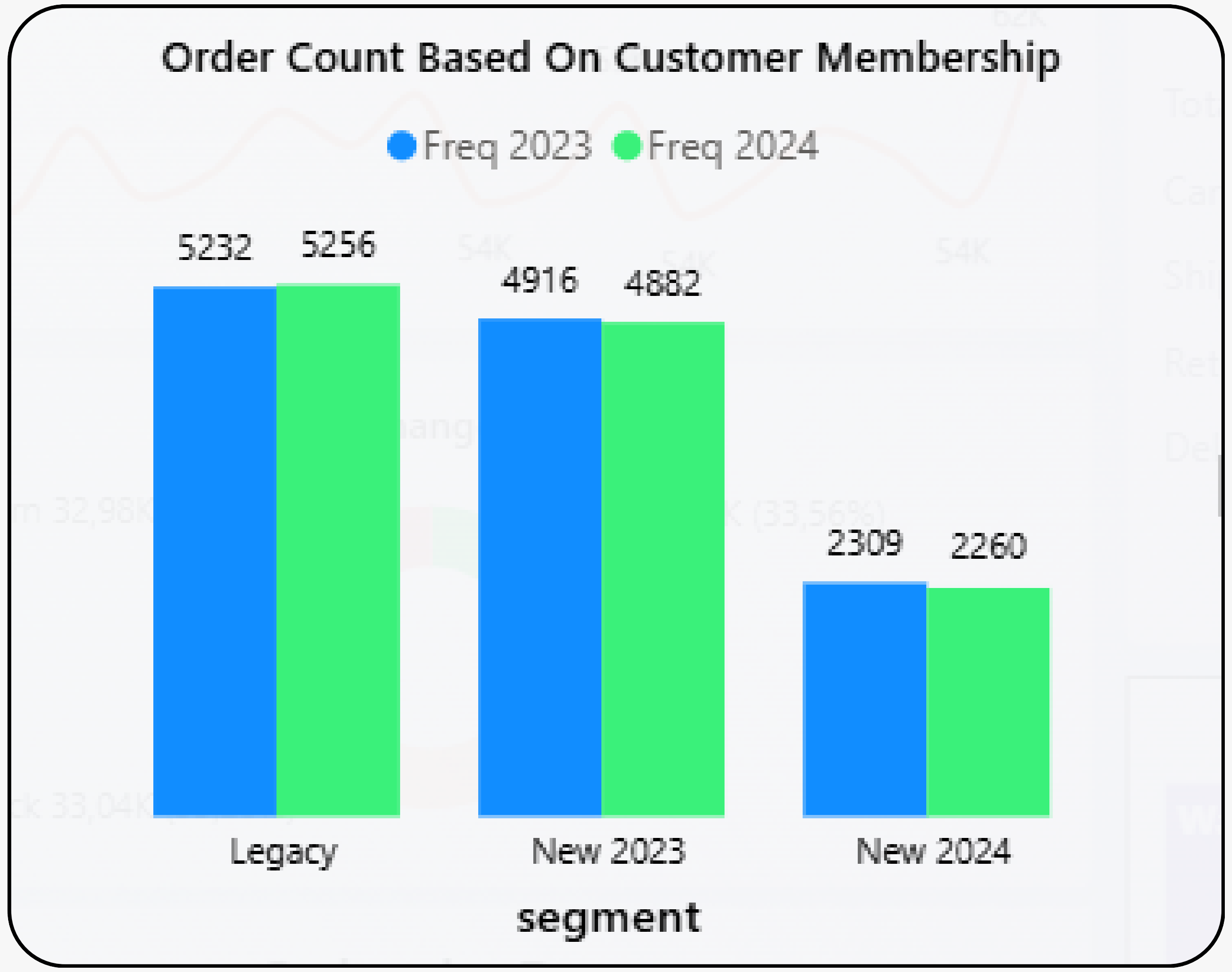
The current inventory metrics reveal a critical **imbalance between stock levels and market demand**. With an Average Stock to Sales Ratio of 515.37, the **business is carrying over 500 times the inventory required** to meet monthly sales, indicating massive overstocking. This inefficiency is further highlighted by a very low Average Turnover Index of 0.26, where nearly **half of the product catalog moves at a rate below 0.25**. This "frozen capital" increases holding costs and the risk of dead stock, as products sit in warehouses for extended periods without converting into revenue.

CUSTOMER AND SALES

STABLE LOYALTY AMIDST SLOWING NEW MEMBER GROWTH

Total Customer

LEGACY	Customer who sign up in 2022 (initial customer)	**2085**
NEW 2023	Customer who sign up in 2023	**1975**
NEW 2024	Customer who sign up in 2024 (new customer)	**915**



The company's revenue remains anchored by highly **loyal Legacy and 2023 cohorts**, who have maintained consistent transaction frequencies year-over-year. However, **a significant 53% drop in 2024** sign-ups indicates a cooling acquisition phase. While the "stickiness" of established members provides a reliable foundation, the slowing influx of new customers highlights **an urgent need to refresh marketing strategies** and diversify the acquisition funnel to ensure future growth.

THANK YOU

2026 2026 2026 2026

Lets work together as you find me!

Where Communication Meets Data.
Growing People, Business, and Impact.

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